NAME - Debapratim Bose
Artificial Intelligence InternsElite
BATCH - May 2026

Fraud Detection System Using Logistic Regression and Transaction Dataset Analysis

INTRODUCTION

Fraud has emerged as one of the key challenges faced by the organizations due to the increasing use of internet and digital payment methods. Early detection of fraud not only reduces financial loss but also increases the security of the financial organization. In this project, we intend to develop a machine learning model for detecting fraud transactions based on their characteristics.

We have used various Python packages, such as Pandas, NumPy, Matplotlib, and Scikit-learn, for developing our project.

PROBLEM STATEMENT

The goal of the project is to build a predictive model that can detect any fraudulent transaction from the transaction data. As fraud is a relatively uncommon activity compared to normal transactions, the model needs to be able to detect any potential fraud without generating too many false positives.

Transaction type, transaction amount, pre-transaction balance, post-transaction balance, isFraud, and isFlaggedFraud are some of the columns present in the dataset.

RESULT AND DISCUSSION

Initially, the project involved an exploration process where the dataset characteristics and transaction trends were determined through visualization techniques. A range of visual representations were

developed to examine the different transaction types, fraud rates by transaction type, and temporal aspects of fraud activity.

The following processes were carried out during data preprocessing:

1. **Removing unnecessary columns including accounts and labeled fraud instances.**
2. **Converting categorical variables into numeric format.**
3. **Normalizing the numeric variables.**
4. **Separating the dataset into training and test datasets.**

The Logistic Regression algorithm was used to train the model with the help of a machine learning pipeline. In order to assess its performance, a classification report with accuracy metrics including precision, recall, and F1-score was generated.

It was discovered that some transactions were more vulnerable to fraud than others. Preprocessing and feature engineering processes enhanced the detection capabilities of the machine learning model. The developed model was stored as Fraud_Detector.pkl.

CONCLUSION

In conclusion, this project was able to effectively show the capabilities of machine learning in the detection of fraud in financial transactions. The application of data analysis and preprocessing, together with the use of Logistic Regression, allows for the identification of possible cases of fraud in financial transactions efficiently. However, the performance of the algorithm could be further improved through the implementation of more complex algorithms like Random Forest, or Neural Network.